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标题	Visible‐Light‐Mediated Heterocycle Functionalization via Geometrically Interrupted [2+2] Cycloaddition
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摘要（原文节选）

AbstractThe [2+2] photocycloaddition is the most valuable and intensively investigated photochemical process. Here we demonstrate that irradiation of N‐acryloyl heterocycles with blue LED light (440 nm) in the presence of an IrIII complex leads to efficient and high yielding fused γ‐lactam formation across a range of substituted heterocycles. Quantum calculations show that the reaction proceeds via cyclization in the triplet excited state to yield a 1,4‐diradical; intersystem crossing leads preferentially to the cl ⋯

数据表 · Table 1（7 行，含数值行 6 行）

Optimization: photocyclization of 2‐acryloyl indole derivative.[a]

Entry[b]	Catalyst	1H NMR yield [%][b]
1	none	0
2	Ru(bipy)3(PF6)2	0
3	[Ir(dtbbpy)(ppy)2]PF6	6
4	fac‐Ir(ppy)3	71
5	[Ir(dF(CF3)ppy)2(dtbbpy)]PF6	81
6	Ir(dFppy)3	92 (85)[d]
7[c]	Ir(dFppy)3	95 (93)[d]

脚注（逐字）：[a] Reaction conditions: 1 (0.1 mmol), catalyst (1 mol.%), 35 W blue LED (440 nm), AcOEt ([1]=0.04 mol dm − 3), r.t., 18 h. [b] 1H NMR yield measured vs. CH2Br2 as internal standard. [c] AcOEt ([1]=0.1 mol dm − 3), addition of NaOAc (10 mol %). [d] Yields in parentheses are for isolated material. bipy=2

正文对应句（逐字）：The resulting pyrrolo[1,2‐a]indole scaffold is an important motif found in numerous natural products. [8] Although photochemically mediated electron‐transfer processes have been used in the past to enable generation of these structures, there are, to our knowledge, no examples employing energy trans

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